

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Valerune — Web-Based Retail Inventory Management System

*Academic Project Specification | IEEE Std 830-1998 Format
B.Tech 3rd Year Capstone Project Submission*

Attribute	Details
Project Title	Valerune — Web-Based Retail Inventory Management System
Student Name	[Student Name]
Roll Number	[Roll Number]
Degree & Program	B.Tech in Computer Science & Engineering
Academic Year	2026–2027
Department	Department of Computer Science & Engineering
Institute	[Institute Name]
Project Guide	[Project Guide Name]
Designation	Assistant Professor / Senior Faculty
Submission Date	October 2026

Document Control & Revision History

This document details the complete functional and non-functional specifications for Valerune. It serves as the official project contract between student, guide, and evaluator.

Version	Date	Author	Changes / Highlights	Status
v1.0	28-Sep-2026	Engineering Team	Initial Architecture Draft & System Requirements	Draft
v1.1	29-Sep-2026	Engineering Team	Added Warehouse Management & Supplier CRUD API Modules	Reviewed
v1.2	30-Sep-2026	Engineering Team	Integrated JWT Security Filter & Order Fulfillment Workflow	Verified
v2.0	01-Oct-2026	Engineering Team	Final Academic SRS Baseline (IEEE Std 830-1998 Compliant)	Approved

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to establish a comprehensive, accurate, and formal definition of the functional and non-functional requirements for Valerune — Web-Based Retail Inventory Management System. It acts as the official reference for project guides, academic evaluators, system architects, and software engineers during the B.Tech 3rd-year capstone review.

1.2 Scope of the System

Valerune is a multi-tier enterprise retail inventory management application designed to bridge customer e-commerce operations with administrative supply chain and multi-warehouse logistics.

In-Scope Functionality: Customer Storefront, Stateless JWT Authentication, Product Catalog CRUD, Multi-Warehouse Inventory Management (4 regional hubs: WH-NORTH, WH-CENTRAL, WH-WEST, WH-SOUTH), Stock Movement Audit Log, Order Fulfillment Workflow, Supplier Directory Management, Operational KPI Analytics Dashboard.

Out-of-Scope Functionality: Real-world credit card payment processing integrations, physical hardware barcode/RFID scanner devices, and 3PL courier tracking API integrations.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Full Expression	Definition / Context
SRS	Software Requirements Specification	Formal document specifying software capabilities according to IEEE Std 830-1998.
API	Application Programming Interface	Set of RESTful HTTP endpoints enabling React frontend & Spring Boot backend communication.
REST	Representational State Transfer	Architectural style for lightweight, stateless Web APIs.
JWT	JSON Web Token	Compact URL-safe token format used for stateless authentication headers.
RBAC	Role-Based Access Control	Security mechanism restricting system access based on user authorization roles (CUSTOMER, ADMIN).
CRUD	Create, Read, Update, Delete	The four fundamental persistence operations for database entities.
SKU	Stock Keeping Unit	Unique alphanumeric product identification code.

1.4 References

1. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
2. Spring Boot 3.3 Reference Manual.
3. React 19 Developer Guide.
4. PostgreSQL 17 Documentation.

1.5 Document Overview

Section 2 outlines system architecture & environment. Section 3 details functional modules. Section 4 specifies external interfaces. Section 5 describes NFRs. Section 6 presents UML analysis models. Section 7 provides academic sign-off tables.

2. Overall Description

2.1 Product Perspective

Valerune operates in a client-server paradigm:

React 19 / TypeScript SPA Frontend <--> REST HTTP/JSON + Bearer JWT <--> Spring Boot 3.3 Backend <--> PostgreSQL 17 Database.

2.2 User Classes and Characteristics

User Class	Technical Proficiency	Primary Responsibilities	Access Rights
Administrator	Moderate to High	Manages product catalog, warehouse stock adjustments, supplier directories, order fulfillment, and auditing.	Full CRUD access across all administrative APIs and /admin frontend routes.
Registered Customer	Basic	Browses product catalog, manages cart/wishlist, places sales orders, and tracks order history.	Access to storefront routes, cart/checkout, personal account profile, and customer order history.
Guest User	Basic	Explores public catalog, views product detail pages, searches items, and filters categories.	Read-only access to public product, warehouse, and supplier informational APIs.

2.3 Operating Environment

- Client: Chrome, Firefox, Edge, Safari browsers.
- Frontend: Node.js 18+ / Vite 8.3 React 19 SPA.
- Backend: Java 21 / Spring Boot 3.3.0 on Port 8080.
- Database: PostgreSQL 17 on TCP Port 5432.

2.4 Design and Implementation Constraints

- B.Tech 3rd-year capstone project scope.
- Database dependency on PostgreSQL 'retail_inventory'.
- Passwords encoded using BCrypt; REST API secured via Bearer JWT.

3. System Features and Functional Requirements

3.1 Authentication & Access Control Module

Req ID	Feature Description	Input / Validation Rule	Priority
FR-1.1	User Registration	Valid name, unique email format, password >= 6 chars. Default role CUSTOMER.	High
FR-1.2	User Login	Valid email and matching password. Returns JWT token.	High
FR-1.3	BCrypt Hashing	Passwords hashed using BCrypt algorithm prior to database persistence.	High
FR-1.4	JWT Generation	Generate signed 512-bit HMAC JWT token upon successful authentication.	High
FR-1.5	RBAC Enforcement	Restrict /admin routes and admin REST endpoints to ADMIN role.	High
FR-1.6	Get Current User	Extract user profile using Bearer JWT token.	Medium
FR-1.7	User Logout	Clear SecurityContext session and client token.	Medium

3.2 Product Catalog Management Module

Req ID	Feature Description	Input / Validation Rule	Priority
FR-2.1	Create Product	Requires unique SKU (VAL-***), name, price > 0, category.	High
FR-2.2	Retrieve Catalog	Returns list of active products with price, rating, image URL.	High
FR-2.3	Product Detail Lookup	Fetches specific product record by UUID string.	High
FR-2.4	Update Product	Modifies product attributes while preserving SKU uniqueness.	Medium
FR-2.5	Soft Delete Product	Sets product active flag to false.	Medium
FR-2.6	Search & Filter	Filters catalog by keyword, category, price range, sorting.	High

3.3 Warehouse & Inventory Management Modules

Req ID	Feature Description	Input / Validation Rule	Priority
FR-3.1	Register Warehouse	Requires unique code (WH-***), name, location, capacity > 0.	High
FR-4.1	Create Stock Mapping	Maps product UUID to warehouse code with quantity >= 0.	High
FR-4.3	Adjust Stock Quantity	Updates on-hand and reserved quantity with non-negative validation.	High
FR-4.4	Reorder Alerting	Triggers warning whenever quantity on hand <= reorder level.	High
FR-4.5	Optimistic Locking	Enforces optimistic locking version control to prevent concurrency conflicts.	High

3.4 Order Fulfillment & Stock Audit Modules

Req ID	Feature Description	Input / Validation Rule	Priority
FR-5.1	Log Movement Event	Records stock transaction type (INBOUND, OUTBOUND, ADJUSTMENT, RESERVATION).	High
FR-6.1	Create Sales Order	Accepts order payload with line items, total calculation, and warehouse code.	High
FR-6.2	Inventory Reservation	Automatically deducts quantity from on-hand and reserves stock.	High
FR-6.4	Update Order Status	Transitions status (PENDING -> PROCESSING -> COMPLETED / CANCELLED).	High

4. External Interface Requirements

- User Interface: Customer Storefront Portal & Admin Operational Dashboard.
- Software Interfaces: React 19 SPA, Spring Boot 3.3 REST API, PostgreSQL 17 DB.
- Communication Interfaces: HTTP/HTTPS REST JSON with Bearer JWT headers.

5. Non-Functional Requirements

NFR ID	Attribute	Target Specification / Standard	Verification Method
NFR-1.1	Performance	REST API response time < 500ms for query endpoints.	Postman & Actuator metrics
NFR-1.2	Security	Passwords hashed using BCrypt. Stateless 512-bit HMAC JWT.	Code audit & security tests
NFR-1.3	Reliability	Database transaction rollback on error. Zero runtime crashes.	JUnit test suite (126 tests)
NFR-1.4	Portability	Cross-browser execution across Chrome, Firefox, Edge, Safari.	Cross-browser verification
NFR-1.5	Maintainability	Layered Spring Boot architecture. Modular React components.	Static code review
NFR-1.6	Usability	Fully responsive layout with inline input validation.	Responsive UI audit

6. System Design and Analysis Models

6.1 Relational Entity Relationship Diagram (3NF)

Entity Name	Primary Key	Foreign Keys	Key Attributes
users	id (UUID)	None	email (Unique), password_hash, name, role (Enum: ADMIN, CUSTOMER), active
products	id (UUID)	None	sku (Unique), name, unit_price, original_price, category, brand, active
warehouses	id (UUID)	None	code (Unique), name, location, capacity, active
inventories	id (UUID)	product_id -> products(id)	warehouse_code, quantity_on_hand, quantity_reserved, reorder_level, version
suppliers	id (UUID)	None	code (Unique), name, contact_name, email, phone, active
orders	id (UUID)	None	order_number (Unique), status (Enum), total_amount, order_type, warehouse_code
order_items	id (UUID)	order_id, product_id	quantity, unit_price, total_price
stock_movements	id (UUID)	inventory_id -> inventories(id)	movement_type (Enum), quantity, reference_number, timestamp

6.2 Sequence & Architecture Models

Authentication Flow: Customer UI -> AuthController -> AuthService -> UserRepository -> PostgreSQL DB -> PasswordEncoder -> JwtTokenProvider -> Returns JWT Bearer Token.

Layered Architecture: React 19 Frontend -> REST API Gateway -> Spring Security Filter -> Controller Layer -> Service Business Logic -> Spring Data JPA / Hibernate ORM -> PostgreSQL 17.

7. Academic Review & Sign-Off

Review Phase	Target Date	Evaluation Parameters	Status / Remarks
Phase I: SRS & Synopsis	October 2026	SRS Completeness, IEEE Std 830 Compliance, Architecture Definition	Completed & Approved
Phase II: Mid-Term Review	December 2026	Database Schema, Authentication Flow, Core Backend API Execution	Pending Review
Phase III: Final Demo	March 2027	End-to-End Integration, Code Quality, Test Coverage, Final Viva	Pending Review

Role	Name & Designation	Signature & Date
Student Developer	[Student Name] Roll No: [Roll Number]	_____ Date: ____/____/2026
Project Guide	[Project Guide Name] Assistant Professor, CSE Dept.	_____ Date: ____/____/2026
HOD / Head of Dept.	Head of Department Computer Science & Engineering	_____ Date: ____/____/2026